

Responses of Plants to Nano-pollution at physiological and molecular level

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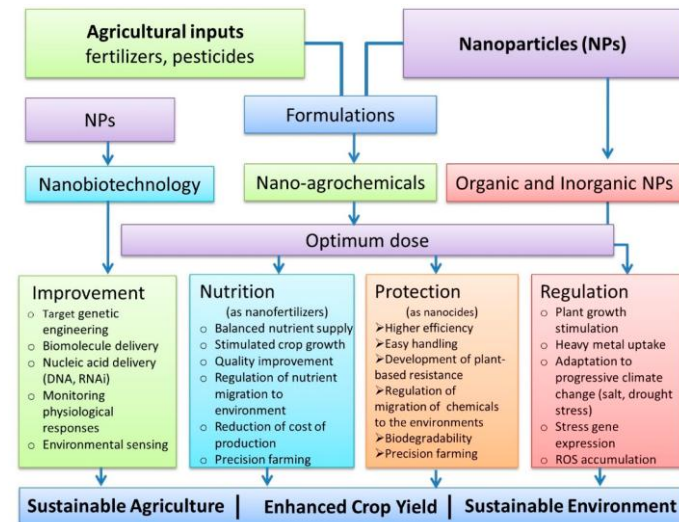


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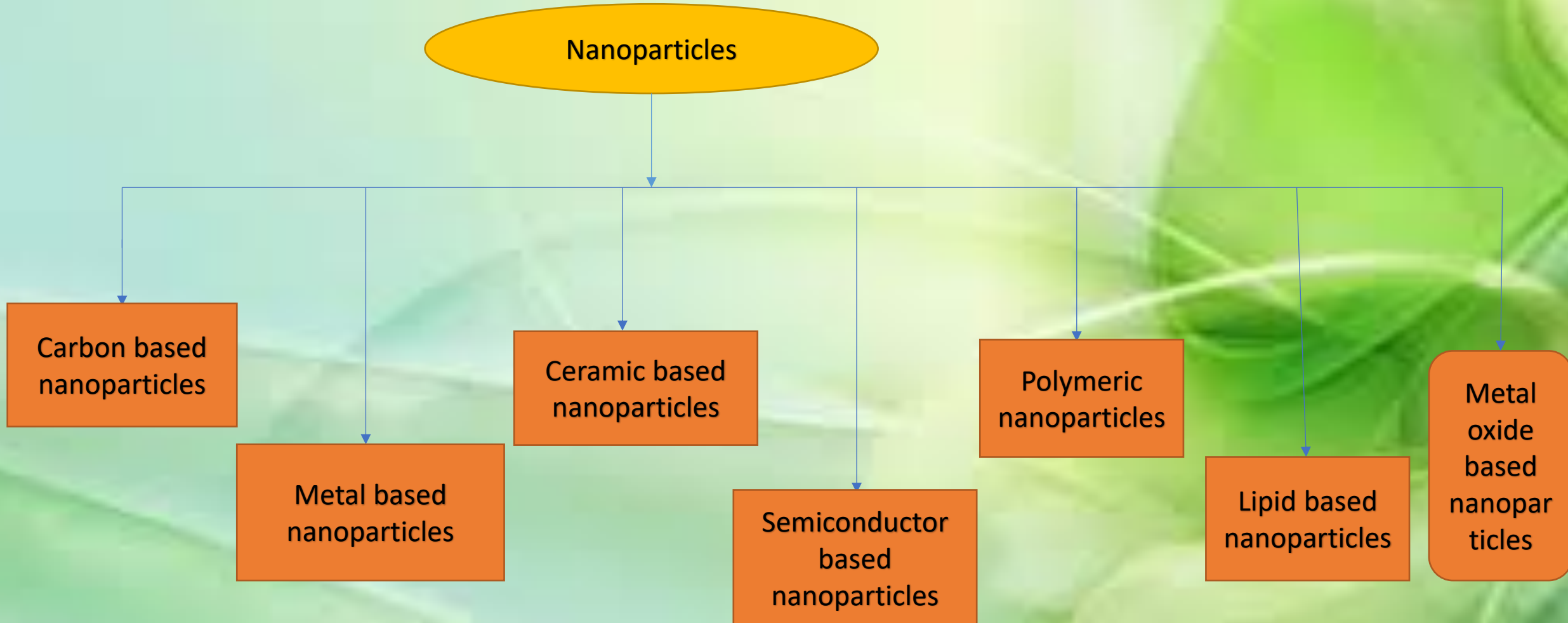
Introduction to Nanobiotechnology and Nanoparticles along with it's use in Plant Science

- Nanobiotechnology is a new field of science that introduces special physicochemical and biological properties of nanostructures and their application in various field such as medicine and agriculture.
- The field of nanobiotechnology has an increased gain in interest in plant science, as the nanomaterials (NMs) has a great potential to enhance crop productivity and it also can be used as vehicles of agrochemicals or biomolecules in plants.
- The NPs (nano particles) are extremely fine particles with lengths between 1 and 100 nm in at least two of their dimensions.
- Nanoparticles have been widely used in many aspects of daily life and energy production, including in catalysts, semiconductors, cosmetics, drug carriers and environmental energy because of their unique properties and novel features.
- Plants plays an important role in transportation of nanoparticles.
- Larger nanoparticles accumulate in the apoplastic space and smaller nanoparticles travel by the symplast process are demonstrated through physiological studies.
- There is a need of evaluation of the toxic effects of nanomaterials in photosynthetic species as a way to assess for their ecological impact.
- Research in this area has been focused in recent times on the interaction between plants and nanoparticles, and the effects of nanoparticles on ecology, the food chain and human health; evaluating the pros and cons of nanoparticles requires interdisciplinary knowledge



Classification of Nanoparticles

NPs are broadly classified into various categories depending on their chemical properties, size and morphology. Some of the well-known classes of NPs are given as below based on their physical and chemical characteristics.



Phytotoxicity in Plants

- Phytotoxicity is simply defined as a plant damage- a toxic effect- from something (e.g. Nanoparticles) the plant was exposed to.
- The phytotoxicity test is a 3-day bioassay based on the seed germination and root growth of the 3 higher plant species after exposure to contaminated soils, solid wastes, composts, sludges or chemicals added to soil.
- Treatment for phytotoxicity in a small area of a plant may involve simply looping off the stem to prevent the damage from interfering with the rest of the plant. Providing adequate water and good general care will usually rally the plant over time and reduce the chance of permanent injury.

There are some methods which are used to proof phytotoxicity in plants when exposed to nanoparticles

- Nanoparticle characterization
- Determination of germination percentage during germination of seeds
- Determination of root/shoot elongation and dry weight during seedling growth
- Mitotic index
- Reactive oxygen species determination
- Hydra acute toxicity test
- Daphnid acute toxicity test
- TEM (transmission electron microscope) observation and Xray microanalysis
- Data analysis

Nanotoxicity in Plants

- ✓ Nanotoxicity studies the toxicity of nanomaterials or nanoparticles.
- ✓ Materials that are toxic at the nanoscale might not be toxic at the microscale.
- ✓ Several researchers have reported both positive and negative effects of nanoparticles on plants, reflecting the variety of plants species, nanoparticle properties, surface coating and culture media.
- ✓ The potential risk from metal based nanoparticles in the environment have increased with the rapidly rising demand for and use of nano enabled consumer products.
- ✓ Plant's central roles in ecosystem function and food chain integrity ensure intimate contact with water and soil system, both of which are considered sinks for nanoparticle accumulation.

At first the nanoparticles are synthesized and characterised then the following test method are used to draw conclusion regarding nanotoxicity

- ✚ Optical measurements
- ✚ Assay treatments
- ✚ MTS cell proliferation/ viability assay
- ✚ Alamar blue cell viability assay
- ✚ Catalase assay
- ✚ Statistics
- ✚ Survey of nanotoxicological papers

Negative impact of excess Nanoparticles on Plant

Plant's response at Physiological level

Disrupts thylakoid membrane

Reduces chlorophyll content

Inhibits photosynthesis and growth of plant

Blockage of water channels through adsorption

Inhibition of elongation of roots

Inhibition of seed germination

Reduces chlorophyll concentration

Inhibits photosynthesis

Elevates the activities of catalase and superoxide dismutase

$\gamma\text{Fe}_2\text{O}_3$ nanoparticles increases IAA and ABA content in the roots of transgenic and non transgenic rice

CeO_2 nanoparticles decreases content of trans- zeatin- riboside

Depletion in growth of tobacco seedlings occurs when exposed to high dose titanium oxide nanoparticles.

Plant's response at Molecular level

Silver nanoparticles at higher dose had induced genes engaged in phenol biosynthesis and glucosinolates

Excess silver nanoparticles has developed disruption of cell division mainly in onion cells and chromosomal aberrations

Silver nanoparticles can bring about formation of chromatin bridges, disturbed metaphase, DNA damage and multiple chromosomal breaks

Higher concentration of silver nanoparticles cause an increase in DNA damage by DNA laddering, comet, and Tunel assays

When tobacco plants are exposed to titanium oxide nanoparticles there is a significant deviation in the expression profiles of miRNAs

Abiotic (salt, water deficit, oxidative) and biotic (defence to pathogens and wounding) stress-reactive genes were induced when exposed to zinc oxide nanoparticles

A concentration-dependent modification in chromosomal aberrations and mitotic activity was observed as an effect of titanium oxide nanoparticles in purple broad bean

When the downregulated genes are exposed to ZNO nanoparticles then it results in biogenesis and cell organization, including, nucleosome assembly, translation and microtubule-based process

Genome-wide genomic and transcriptomic approach can determine the toxic impacts on exposure to cadmium sulphide quantum dots

A huge number of genes are there whose transcript abundance was correlated with the cadmium sulphide quantum dots tolerance like an MYB containing gene ELM2 can be recognised

Single-walled carbon nanotubes are proved to exert adverse effects through oxidative stress on Arabidopsis and rice leaf protoplasts

Carbon nanotubes causes apoptosis or programmed cell death, chromatin condensation and DNA damage

A scientist assessed the molecular and physiological responses is activated when exposed to CeO_2 and indium oxide nanoparticles on Arabidopsis

It was proved that copper oxide nanoparticles can mediate DNA damage in terrestrial plants

Severe alterations in expression patterns of the genes required for root development is induced by the graphene oxide exposure combined with drought and salt stress

Steps to reduce the nano-pollution in plants Application of nanoparticles in plants (Positive Impact)

Detoxification
mechanism in
plants:

Antioxidant
Defence
System –
Nonenzymatic
Agents:

Antioxidant
Defence
System –
Enzymatic
Components:

Highest grain yield can be achieved by applying nano-iron oxide. Some nanoparticles also protect chloroplasts from aging for an extended period of light. .

When the bulbs are soaked in a silver nanoparticle solution, NPs helps in proper elongation and flowering in plants and proved to be an effective strategy to promote growth in plants

Alumina nanoparticles have proved to cause an increase in number of fronds, root length and overall biomass in *Lemna minor*. Higher number of flowers can be produced when it is plants are treated with silver nanoparticle solution and remain flowered for longer period.

When aged spinach seeds are soaked in high concentration of titanium nanoparticles solution then the germination rate increases. Treating with titanium nanoparticles can accelerates nitrogen assimilation and promotes the growth of spinach

The intense use of nanoparticles in modern age has improved the life style of humans in greater extent. Environment cannot be neglected as it is needed in present and modern life. It is proved from the studies that metal oxide and metal nanoparticles are dangerous to plants, when used in excess whereas, when present in minor quantity it can be useful for plants. Hence a sharp increase in the concentration of nanoparticles in the environment may cause an adverse effect on agriculture in later life. A clear knowledge can be obtained from the review to prove the effect of metal oxide nano particles on the plants but the interaction of plants with nanoparticles at a molecular is not understood at a large extent, and the studies are conducted in its primitive stage. A lot of research is needed before bringing the nanoparticles to the field. Morphological variation caused to plants due to metal oxide and metal nanoparticles is studied in most of the researches. Research is needed to be performed to understand the impact of metal oxides and metal nanoparticle on plant's molecular mechanism and its physiology.

Researches needed to be done for the awareness of the people about the proper disposal of nanoparticle products. In the morphological field of plant science there is lack of standardisation of the phytotoxic assay and thus studies should be made to improve it. Research is also needed to be done for eradication of nanoparticle from agriculture soil and wastewater to save the environment from getting damaged.

Conclusion & Future Perspectives

*Thank
you*

